

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2023

Course Code: MEC101

Course: **BASIC MECHANICAL ENGINEERING**

QP No. :U007R-1

Duration: 3 hours

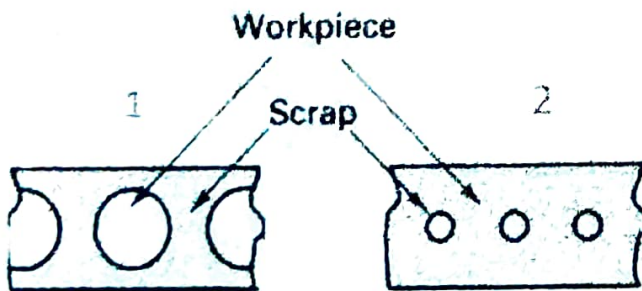
Max. Marks:100

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. In a cylindrical job, a 16 mm finely finished hole is to be made, list out the operations need to be performed in the lathe machine.
2. Identify the following sheet metal process and differentiate them.



3. To convert cylindrical rod into square rod, which is made up of cast iron, whether hot or cold forging is preferable? Why?
4. To compensate shrinkage loss, what provisions should be made in sand casting?
5. What is carburizing flame in oxyacetylene welding?
6. If a crankshaft revolves 4 times, how many power strokes will be obtained in four stroke and two stroke engines respectively.

7. What type of ignition system is preferred in racing cars? Why?
8. In a hot and humid environment like Chennai, what type of air conditioning system is preferred?
9. State the operational difference between steam turbine and gas turbine.
10. In a hydroelectric power plant, to stop the sudden pressure rise what device should be installed?

PART - B

Answer all the questions

4 x 15 = 60 Marks

11. Enumerate any 5 operations which can be performed on drilling machines with a neat diagram.

(OR)

12. Explain the principle of rolling and with a neat sketch explain any 4 types of rolling mills.

13. Describe the working principle of manual metal arc welding with a neat diagram and differentiate A.C. welding and D.C. welding.

(OR)

14. Describe the steps involved in making green sand molds, draw a suitable sketch and show the important parts.

15. With a neat sketch of Vapor compression refrigeration system, explain its component parts and working principle.

(OR)

16. Explain the working principle of fuel injection system for diesel engine with a neat sketch of fuel pump and injector.
17. What is high pressure boiler? Explain any one type of high-pressure boiler with a neat sketch.

OR

18. Explain the working principle of nuclear power plant and its component with a neat sketch.

PART - C

Answer the following

1 x 20 = 20 Marks

19. The connecting rod of the engine swings about the crosshead pin in 2 stroke engines and about the gudgeon pin in 4 stroke engines
- (i) Identify and explain a suitable forging process to manufacture the connecting rod. (10)
- (ii) Differentiate four stroke and two stroke engines. (10)
